Chapter 7

Evaluation of the MVA-HLT and Complementary Offline Selection

7.1 Measurement of Trigger Rate and Timing

[I need an academic source for this section – ask Marco?] As discussed in chapter 5, the first main consideration for any HLT is the trigger rate, usually measured in Hz. There are two main corresponding numbers: the total rate measures the total number of events being accepted by the trigger, whereas the pure rate measures the number of events being accepted by a trigger that would not be accepted by any other trigger. In addition to the rate of accepted data, the overall timing of the trigger is of vital importance. Given the speed with which events are read out from the L1 triggers, every milisecond of computation time is a vital resource to be spent sparingly.

Standardized tools to evaluate the timing and rate of the HLT paths exist (REF?). Firstly, for the timing, special L1 skim files exist which are earmarked for trigger rate measurement. Existing software can be run that measures both the total and pure rates of any selection of triggers. This software returned a total rate of 61.53 Hz of events for the sHLT. The MVA-HLT, on the other hand, yielded a total rate of 29.43 Hz. This measurement is quite in line with the preliminary results shown in section 5.4. This 52.2% relative reduction in trigger rate will be further examined in section 7.2.1. Additionally, the pure rate of the MVA-HLT was measured to be 8.03 Hz. This represents a relatively small addition of events in the overall scale of the trigger rate, and is therefore acceptable. In fact, some degree of pure rate is to be expected given the lowered p_T thresholds for this trigger.

As with the trigger rate, standardized tools exist to measure the HLT path timing (REF?). This measurement runs over a small number of data events and measures the path timing for the whole trigger menu simultaneously. The sHLT, in this framework, takes approximately 15.8ms of processing time. In comparison, the MVA-HLT takes about 15.5ms. While this software doesn't provide uncertainties on these measurements, repeated runs show that the resulting timing values may vary by a few tenths of a milisecond. So, then, it appears that these path timings are relatively consistent with each other. Overall, the important fact is that the MVA-HLT does not take any unduly large amount of processing time; given the application of the BDT within the path, this was a major consideration during the development of the trigger. Additionally, similar software can run to measure the total event throughput per second of the entire HLT menu. This measurement can be run with and without the presence of the MVA-HLT. Running the full menu yields a throughput of 526.2 ± 2.7 events/second. Running with the MVA-HLT added in yields a throughput of 524.8 ± 3.1 events/second. These measurements are clearly within uncertainties of each other, and it can therefore be concluded that the presence of the MVA-HLT does not measurably impact the overall timing of the trigger menu.

7.2 Evaluation of MVA-HLT Eficiency on Selected Signal Samples

Once it has been verified that the trigger rate is within the limits posed by the collaboration, the next major task is ensuring that the signal efficiency for the MVA-HLT is a noticable improvement over the standard HLT. Since the reconstruction, calibrations, and corrections are different at the NanoAOD level, the measurements made at the RAW level in section 5.3 must be verified on independent analysis-level (NanoAOD) samples. HiggsDNA was used to produce these samples and plots, which is described in more detail in 3.4.4. One major point of note is that here the ”All Events” curve shown must be carefully defined. For these MVA-HLT plots, there will be no additional preselection. Only minimal kinematic cuts of $p_T > 14.25$, $|\eta_{SC}| < 2.5$, and $100 < M_{\gamma\gamma} < 180$ are applied. Additionally, both the signal and data are required to pass one or both of the triggers of interest. So, every event replicated below will pass either the sHLT, MVA-HLT, or both.

As in section 5.3, plots can be made of selected input variables and the resulting accepted distributions can be compared for the MVA-HLT versus the standard diphoton HLT. For these susequent plots, the acceptance of MVA-HLT will be shown in dark blue, while the acceptance of the standard HLT will be shown in dark blue. Additionally, the light blue curve will show events that pass the MVA-HLT but fail the standard HLT. Therefore, the light blue curve will our signal gain directly.

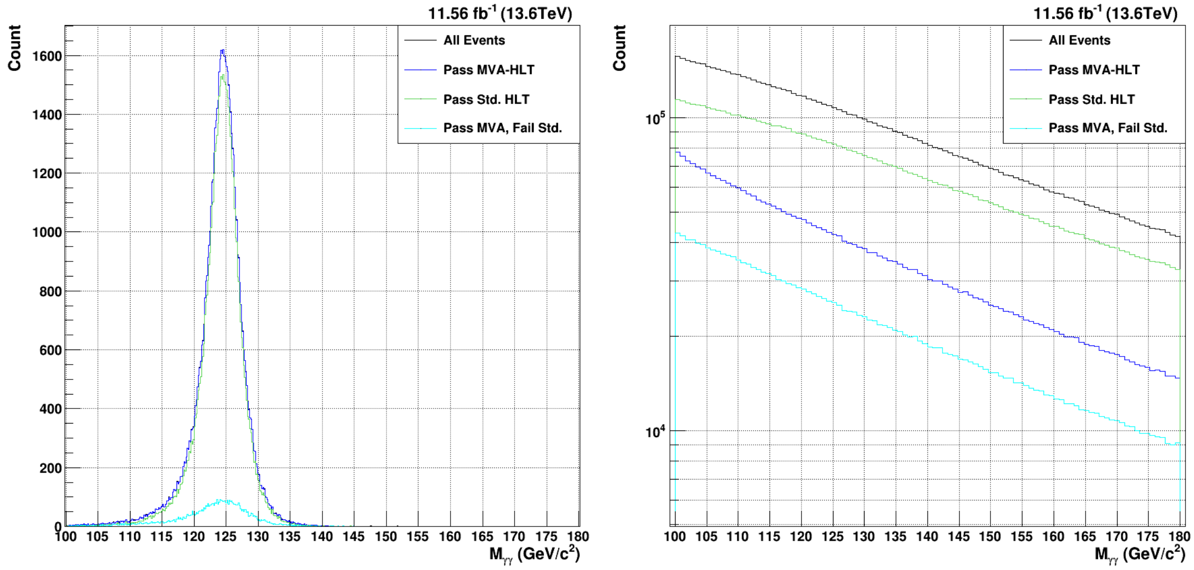
In order to assess these results in terms of real physics yeilds, the counts of signal and data must be luminosity-normalized. The EGamma dataset for the 2024I run consists of a total of 11.56 fb^{-1} of integrated luminosity. In contrast, monte carlo signal samples are produced with a specific number of events in mind. Then, an ”equivalent luminosity” is calculated, essentially measuring how much real integrated luminosity would be required to capture the generated number of events. Additionally, each MC has an associated cross-section (XS) given in pb. When combining this with the event-by-event weights calculated at the generator, the number of expected events can be constructed as given in equation 7.1. The ratio at the end of the equation, (Target Lumi/Equivalent Lumi) is added so that the MC counts are scaled to the same luminosity as the selected data.

$$ExpectedCount = EventWeight * XS(pb) * EquivLumi(fb^{-1}) * 1000 * \left(\frac{TargetLumi(fb^{-1})}{EquivLumi(fb^{-1})}\right) \tag{7.1}$$

7.2.1 $H \to \gamma\gamma$ MVA-HLT Acceptance

Beginning with the $M_{\gamma\gamma}$ distribution (fig. 7.1), it can be seen that the MVA-HLT (blue) has considerably improved signal MC acceptance compared to the standard HLT. In fact, there is a 9.7% relative signal efficiency gain (97.32% vs 88.68%) when considering both the barrel and endcap events together. This is a bit lower than the 13.6% signal gain seen at the RAW trigger level (93.86% vs 82.64%). This is partly due to the improvement in overall signal efficiency for the standard HLT at the NanoAOD level. Because the NanoAOD reconnstruction is imrpoved relative to that at the trigger RAW level, it is expected that the total signal efficiency at this level will be improved. Overall, however, it can be seen that the $H \to \gamma\gamma$ signal eficiency is very high for the MVA-HLT, as desired.

Another major advantage of the MVA-HLT is the reduction in data rate. As demonstrated in section 7.1, the MVA-HLT has approximately half the trigger rate of the standard trigger. This is approximately confirmed by data distribution shown in 7.1b. For all η_{SC}, the total data acceptance is 22.41% compared to thr 42.62% of the standard HLT. The light blue curve, additionally, can be used as a proxy for seeing the pure rate measured in section 7.1; about 13.4% of the data passes the MVA-HLT but fails the standard HLT.

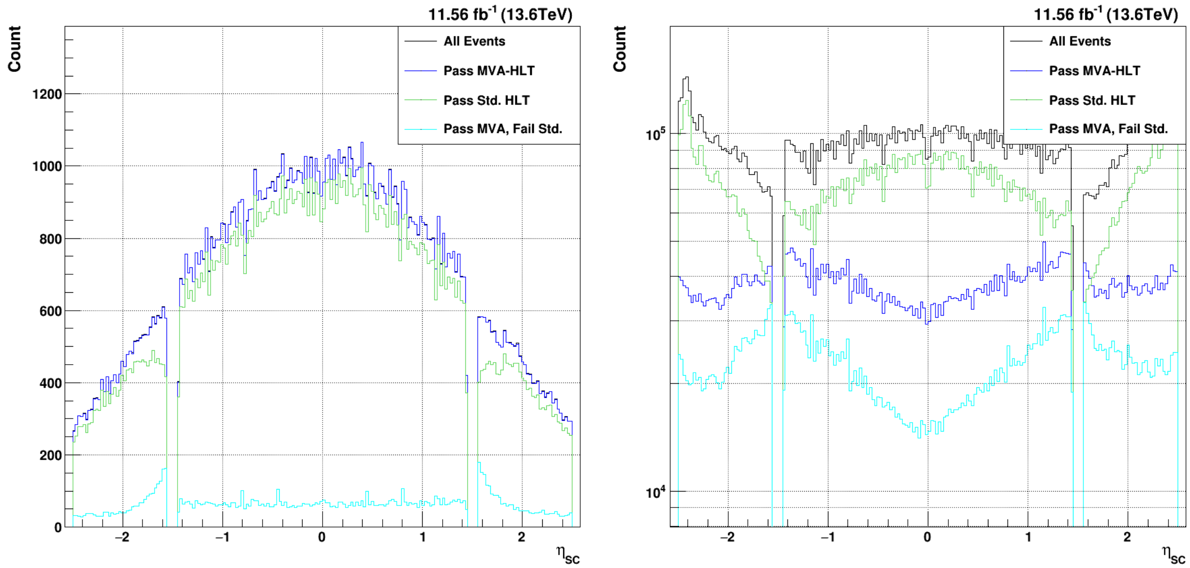


(a) Signal MC $M_{\gamma\gamma}$ **(b)** Data $M_{\gamma\gamma}$

Figure 7.1. $H \to \gamma\gamma$ acceptance of the MVA-HLT (dark blue) vs. the current standard diphoton HLT (green) for $M_{\gamma\gamma}$. Events that pass the MVA-HLT but fail the standard HLT can be seen in light blue.

Next, the acceptance in the trigger in η_{SC} 7.2 can be considered. The MVA-HLT has relatively consistent acceptance in η_{SC}, whereas the standard trigger shows holes in the acceptance in the lowest-η regions of the endcap. Investigations have shown that the R_9 cuts of the standard trigger are responsible for this inefficiency. As seen in the light blue curve, the MVA-HLT is able to close this gap and provide more consistent kinematic coverage. Otherwise, the gain relative to the standard MVA-HLT is relatively flat throughout the barrel and the rest of the endcap. These features shown that the kinematic coverage is improved relative to that of the standard HLT.

The acceptance of the data, shown in figure 7.2b, shows a few relevant features. Whereas the standard HLT accepts more data in the center of the barrel and cuts out more of the barrel above $|\eta_{SC}| > 1.0$. The MVA-HLT, on the other hand, exhibits the opposite features. In the data, the standard HLT closely matches the shape of the full distribution in black, including considerable acceptance in the high-η horns. The MVA-HLT is relatively flat throughout the enecap, though it does also accept a small amount more data in the high-η horns.

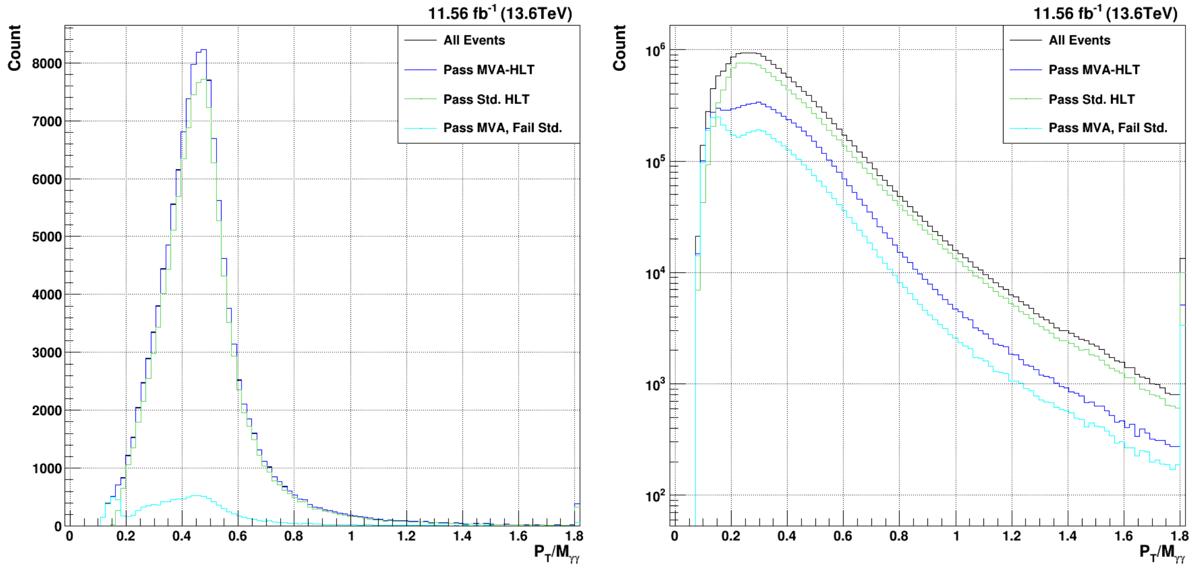


(a) Signal MC η_{SC}

(b) Data η_{SC}

Figure 7.2. $H \to \gamma\gamma$ acceptance of the MVA-HLT (dark blue) vs. the current standard diphoton HLT (green) for η_{SC}. Both the lead and sublead photons are plotted together. Events that pass the MVA-HLT but fail the standard HLT can be seen in light blue.

Next, the photon candidate $p_T/M_{\gamma\gamma}$ can be examined. Given the lower p_T thresholds of the MVA-HLT, it is expected that there will be some enhancement in acceptance for low-p_T candidates. For the sHLT, the signal MC in figure 7.3a goes to 0 around $p_T/M_{\gamma\gamma} = 0.15$, which corresponds to the sublead $p_T > 25$ cut. This leads to an enhancement in the gain below this value, as the MVA-HLT keeps consistent kinematic coverage down to the lowerst values plotted. Of course, these low-$p_T/M_{\gamma\gamma}$ sublead photons correspond to a higher-$p_T/M_{\gamma\gamma}$ lead photon, which is usually not cut out by the sHLT. Since the HLT is a per-event decision, both of these photons will be cut out by the sHLT. So, gains across the whole $p_T/M_{\gamma\gamma}$ range can be seen for the MVA-HLT. For the background, it can be seen that the sHLT begins to fall off at lower values. The MVA-HLT achieves considerably lower data acceptance without explicity cutting out any region of the $p_T/M_{\gamma\gamma}$ distribution.



(a) Signal MC Diphoton p_T

(b) Data Diphoton p_T

Figure 7.3. $H \to \gamma\gamma$ acceptance of the MVA-HLT (dark blue) vs. the current standard diphoton HLT (green) for p_T. Both the lead and sublead photons are plotted together. Events that pass the MVA-HLT but fail the standard HLT can be seen in light blue.

Finally, the results of the signal MC efficiency and the data acceptance are tabulated in table 7.1. The different η_{SC} configurations are separated here – B-B and E-E events correspond to both photons being located in the barrel and endcap, respectively, while the B-E events have one photon in the barrel and the other in the endcap. Immediately, it can be seen that the MVA-HLT has relatively constant signal acceptance for each region, varrying by a total of 3.45% between the B-E events (95.2%) and the B-B events (93.39%). Meanhile, the standard HLT has considerably reduced signal efficiency in the B-E. This allows the MVA-HLT to have a relative efficency gain of 17.1% in this region. This is another way in which the kinematic coverage of the MVA-HLT is greatly improved relative to the standard HLT.

Overall, it can be seen that the MVA-HLT achieves the desired result at the NanoAOD analysis level. The signal efficiency is about 9.7% higher for the $H \to \gamma\gamma$ signal MC relative to the standard HLT. The overall efficiency sits at 97.3%, which allows virtually all of the signal MC to be considered for subsequent analyses. In addition, no holes in kinematic coverage can be

Table 7.1. $H \to \gamma\gamma$ signal efficiencies and background acceptances for both triggers, broken down by the location of each photon candidate in η_{sc}.

(a) Signal Efficiencies

Region	MVA-HLT(%)	Std. HLT(%)
B-B	98.65	93.39
B-E	95.20	81.27
E-E	96.70	86.24
All	97.32	88.68

(b) Data Acceptances

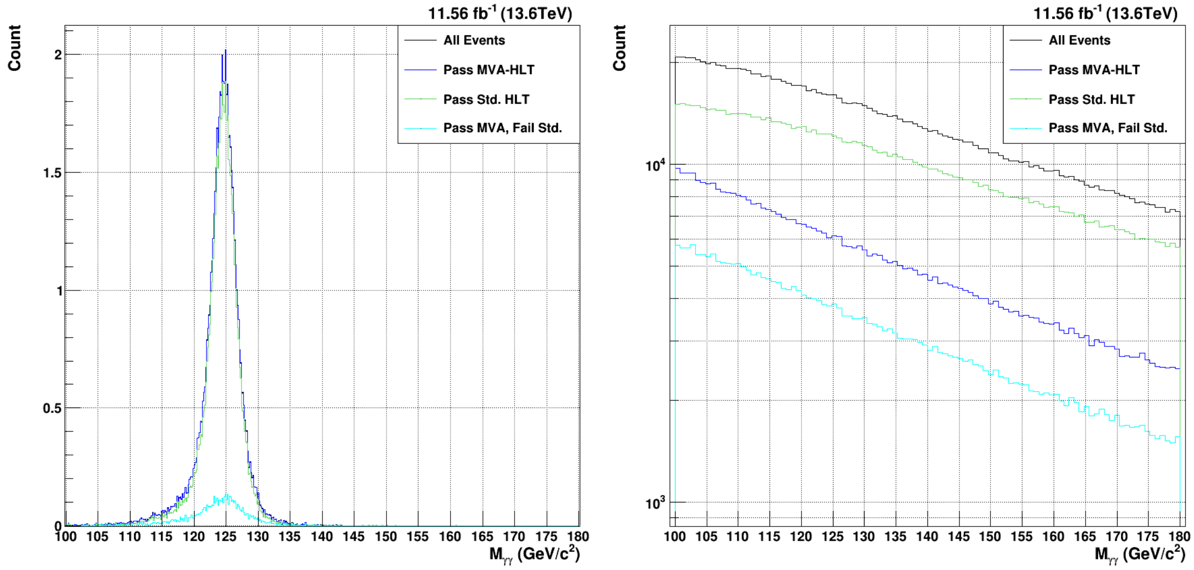
Region	MVA-HLT(%)	Std. HLT(%)
B-B	21.07	37.27
B-E	23.11	46.57
E-E	22.26	32.32
All	22.41	42.62

seen, meaning that the MVA-HLT is able to more-completely cover the full range of possible physics that can be gleaned from $H \to \gamma\gamma$ processes.

7.2.2 $HH \to b\bar{b}\gamma\gamma$ MVA-HLT Acceptance

Where the MVA-HLT can sine best, however, is for more rare processes like the $HH \to b\bar{b}\gamma\gamma$. As with the single Higgs results, plots can be made comparing the signal efficiency and data acceptance for the sHLT versus the MVA-HLT. These distributions are shown in figures 7.4-7.6. Given that the cross section of the DiHiggs is much lower than that of the single Higgs (as described in section 1.4), the equivalent counts for this signal MC are lower.

Similarly to the MVA-HLT results for the $H \to \gamma\gamma$ signal MC, very high signal efficiency is immediately seen for $HH \to b\bar{b}\gamma\gamma$. In figure 7.4, it can be seen that the signal acceptance has near-perfect kinematic coverage in $M_{\gamma\gamma}$ and achieves a total efficiency of 98.45%. Compared to the sHLT (88.45%), this is an 11.3% relative gain. This is, in fact, slightly higher than the 97.32% achieved for the single Higgs. Since the MVA-HLT is largely aimed at improving the signal efficiency for rarer processes, this is a welcome result. The data acceptance is lower than that for the single Higgs, but the ratio between the sHLT and the MVA-HLT is approximately constant – while the sHLT accepts 33.2% of the data, the MVA-HLT accepts 16.86%. There is no particular $M_{\gamma\gamma}$ dependence to the acceptance of either HLT for the signal or background.

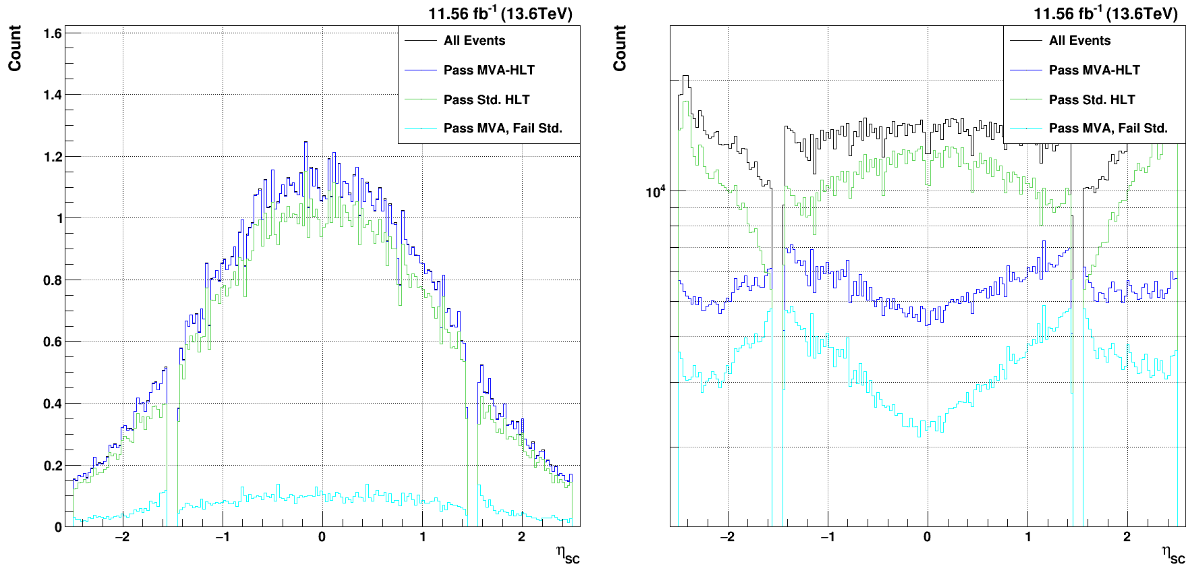


(a) Signal MC $M_{\gamma\gamma}$

(b) Data $M_{\gamma\gamma}$

Figure 7.4. $HH \to b\bar{b}\gamma\gamma$ acceptance of the MVA-HLT (dark blue) vs. the current standard diphoton HLT (green) for $M_{\gamma\gamma}$. Events that pass the MVA-HLT but fail the standard HLT can be seen in light blue.

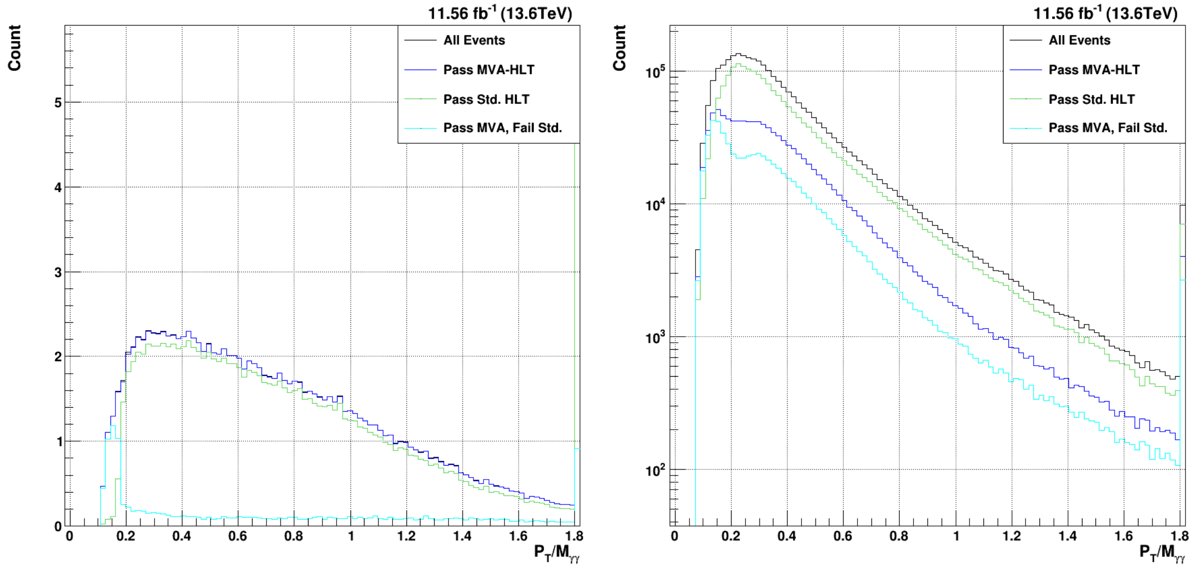
Upon examining the η_{SC} distributions, the kinematic differences between the single Higgs and diHiggs are again borne out; the $HH \to b\bar{b}\gamma\gamma$ signal MC tends to have far fewer events in the endcap and is quite central near 0 for the barrel. The same gains in signal efficiency around the low-η endcap can be seen, though they are smaller for the DiHiggs. The shape of the background acceptance is essentially unchanged, though the dip at $\eta = 0$ is perhaps marginally larger. Again, this reaffirms the fact that the kinematic coverage in η_{SC} is improved for the signal MC relative to the MVA-HLT.



(a) Signal MC η_{SC} **(b)** Data η_{SC}

Figure 7.5. $HH \rightarrow b\bar{b}\gamma\gamma$ acceptance of the MVA-HLT (dark blue) vs. the current standard diphoton HLT (green) for η_{SC}. Events that pass the MVA-HLT but fail the standard HLT can be seen in light blue.

As seen in figure 7.5a, the diHiggs signal MC shows some clear kinematic differences compared to the single Higgs MC. Whereas the single Higgs drops off much faster in $p_T/M_{\gamma\gamma}$, the diHiggs distribution is much broader and extends to higher values. In fact, the overflow bin in the distribution has a very high population, showing that the distribution continues on for quite some time past the bounds of the graph shown. Like the single Higgs, however, the sHLT cuts out the low-$p_T/M_{\gamma\gamma}$ values. The MVA-HLT keeps virtually all of these photons, meaning that there is considerable gain in signal efficiency in this region. The behavior of the data is much the same as that of the single Higgs.



(a) Signal MC $p_T/M_{\gamma\gamma}$ **(b)** Data $p_T/M_{\gamma\gamma}$

Figure 7.6. $HH \to b\bar{b}\gamma\gamma$ acceptance of the MVA-HLT (dark blue) vs. the current standard diphoton HLT (green) for $p_T/M_{\gamma\gamma}$. Events that pass the MVA-HLT but fail the standard HLT can be seen in light blue.

Table 7.2 shows the total efficiencies for both the $HH \to b\bar{b}\gamma\gamma$ signal MC and data in the different detector regions. While the signal MC efficiencies for the sHLT are approximately equal to those of the single Higgs, the MVA-HLT achieves marginally high efficiencies in all of the detector regions relative to the single Higgs. The relative gain, as a result, is higher for the diHiggs MVA0-HLT. Meanwhile, the data acceptance of both HLTs is lower than that of the single Higgs, though the relative proportion stays consistent.

Table 7.2. $HH \to b\bar{b}\gamma\gamma$ signal efficiencies and background acceptances for both triggers, broken down by the location of each photon candidate in η_{sc}.

(a) Signal Efficiencies

Region	MVA-HLT(%)	Std. HLT(%)
B-B	98.90	91.01
B-E	97.25	81.10
E-E	98.26	89.18
All	98.45	88.45

(b) Data Acceptances

Region	MVA-HLT(%)	Std. HLT(%)
B-B	14.73	28.12
B-E	18.26	37.24
E-E	15.51	24.77
All	16.86	33.24

7.3 Evaluation of Offline Selection Eficiency with MVA-HLT

As discussed in chapter 6, the offline selection cuts are designed to cover the corresponding HLT cuts. Since the sHLT is based on hard cuts on the p_T, showershape, and isolation variables, the offline selection is created using similar principles. The offline selection BDTs developed in chapter 6 are, on the other hand, designed to cover the MVA-HLT. While the single photon preselection cuts on single Higgs and diHiggs samples were discussed in sections 6.2.2 and 6.2.4, respectively, the cuts on the corresponding diphoton BDTs were not investigated. The results of these final selections – applied on top of the corresponding HLT – will be displayed in this section.

Any chosen offline selection cuts are only given as a point of comparison; the power of a BDT-based approach lies in its ability to be easily tuned for each analysis. As a result, multiple points of comparison will be shown in subsequent plots for both single Higgs and diHiggs samples. Given that this offline selection must compare to that of the standard analysis, this provides a reasonable starting point. So, then, the first diphton BDT cuts will be chosen such that the overall background acceptance of the BDT-based analysis matches that of the standard anylsis. With the same background rate, the improvement in signal efficiency can then be compared as a direct appraisal of the power of the BDT-based analysis. Additionally, selections with one half and twice as much background as the standard analysis will be shown in order to provide additional points of comparison. In sections 7.3.1 and 7.3.2 selected variable plots will be displayed for $H \to \gamma\gamma$ and $HH \to b\bar{b}\gamma\gamma$, respectively.

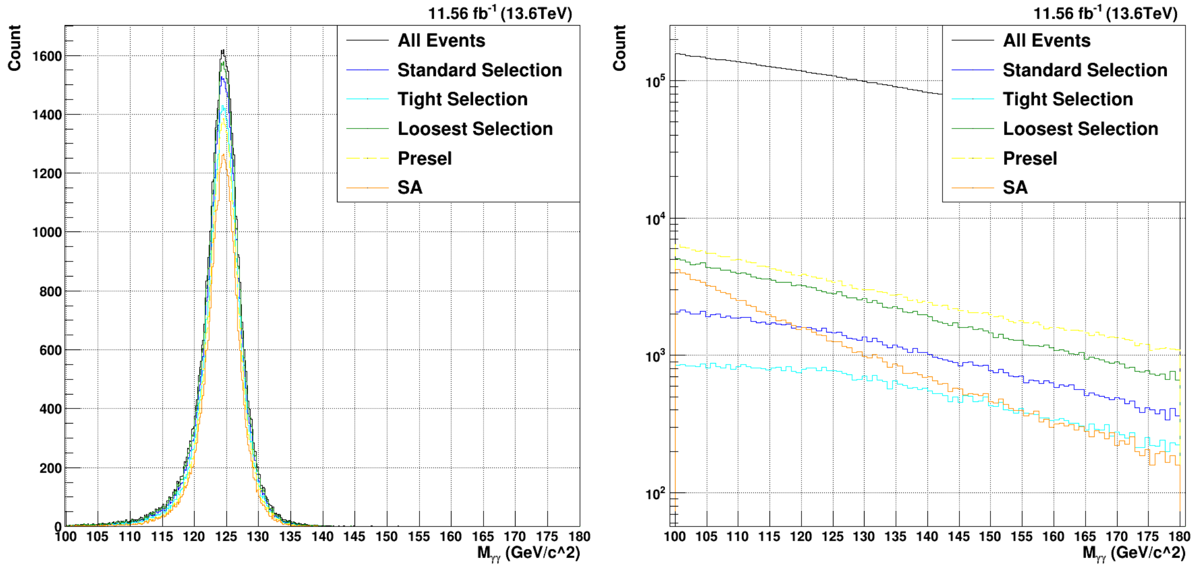
For both the single and diHiggs samples, the plots will be shown in much the same way. The same event scaling described in equation 7.1 will be used to display the expected counts for the signal MC. The plots will contain six curves total: the black curve will show all reconstructed events selected by the basic selection criteria described before; the dark blue, light blue, and light green will show the standard, tight, and loosest BDT-based selections, respectively; the yellow and orange curves will show the single photon preselection and the overall standard analysis cuts, respctively.

7.3.1 $H \rightarrow \gamma\gamma$ Offline Selection Acceptance

The $M_{\gamma\gamma}$, shown in figure 7.7, contains all of the curves described above. Beginning with the signal in figure 7.7a, the improved signal efficiency of all three BDT-based analyses is immediately manifest. Whereas the standard analysis (SA) cuts only achieve a signal efficiency of 70.53%, the diphoton BDT cuts that accept the same amount of backround are able to attain a 91.27% signal efficiency. So, then, the BDT-based selection is not only able to leverage the improved signal performance of the MVA-HLT, but it is able to gain a relative 29.4% improvement in overall signal efficiency. Even the tight selection, which has half the background acceptance of the SA, reaches a signal efficiency of 83.51%. The loosest selection, with twice the background of the SA, has a very high signal efficiency of 95.74%. This selection could be useful for very well-tagged analyses that have some additional criteria that allow the background rate to be brought down considerably without the need for a tight $H \rightarrow \gamma\gamma$ selection.

The shape of the signal mass distribution is, of course, worth investigating. Clearly, with $> 90\%$ signal efficiency, the standard BDT-based selection does little to shape the signal MC mass distribution. While the peak of the distribution has very high efficiency, the high- and low-mass tails are also well-preserved. The SA cuts lose considerably signal efficiency in the center of the distribution near $125 GeV/c^2$. There is also a clear reduction in the low-mass tail with the standard selection. As previously discussed, the measure $2\sigma_{Eff.}$ is useful for comparing the overall width of different mass distributions. A lower $2\sigma_{Eff.}$ corresponds to a narrower mass distribution, with often means that the quality of the preserved events in higher. The full distribution measures at $2\sigma_{Eff.} = 5.81 GeV/c^2$. The standard BDT-based selecton cuts out proportionally more of the tail than the center of the distribution, leading to $2\sigma_{Eff.} = 5.58 GeV/c^2$. The SA, on the other hand, cuts out so much of the tails that it achieves $2\sigma_{Eff.} = 5.17 GeV/c^2$. Of course, while the SA cuts are able to reduce the mass width considerably, this comes with a nearly 30% loss in the signal. While the less rare gluon fusion $H \to \gamma\gamma$ process faces no major derth of events, this major increase in overall signal efficiency is likely worth an increased mass width for rarer processes.

The background mass distribution in figure 7.7b also shows a noteworthy difference between the SA and the BDT-Based selections. Firstly, it can be seen that the preselection curve (in yellow) is relatively parallel to the all curve (in black). The SA curve, on the other hand, is considerably steeper; since the only difference between the preselection and SA cuts are the $p_T/M_{\gamma\gamma}$ cuts, it is clear that these cuts are biasing the mass distribution towards lower masses and removing more high-mass background events. This makes sense, as increasing $M_{\gamma\gamma}$ for the same p_T will of course decrease the $p_T/M_{\gamma\gamma}$ and therefore make an event more likely to fail the cuts. While the BDT-based cuts do not show this behavior, they have reversed relationship with $M_{\gamma\gamma}$. Beginning with the loosest cuts in green, the shape differs little from that of the preselection. As the diphoton BDT cuts tighten, however, it is clear that the low-mass region (below $M_{\gamma\gamma} = 125 GeV/c^2$) is cut out considerably.

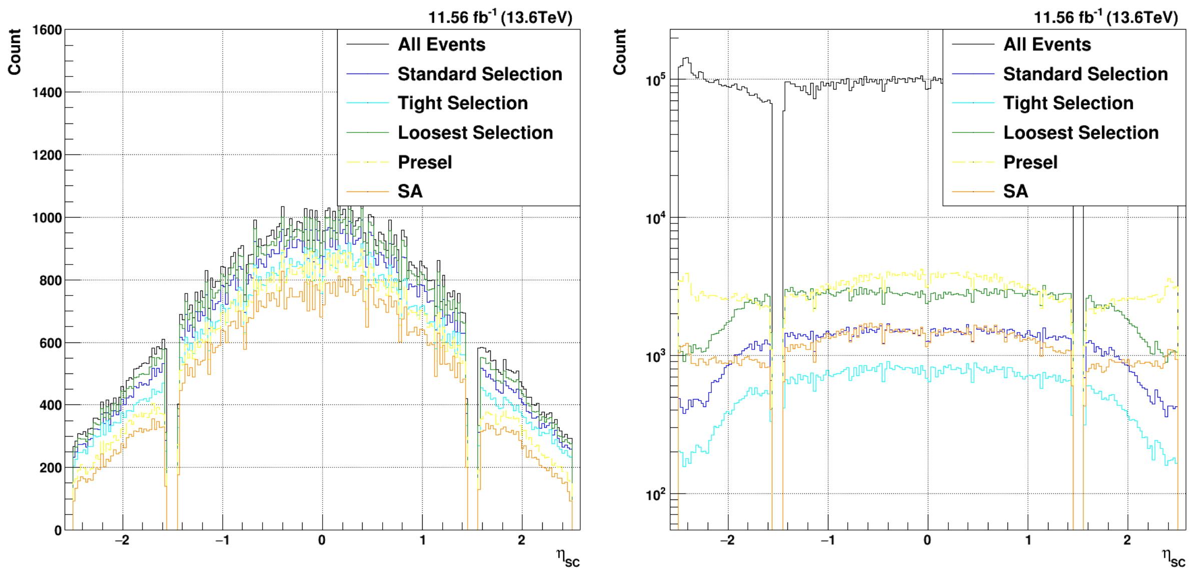


(a) Signal MC $M_{\gamma\gamma}$ **(b)** Data $M_{\gamma\gamma}$

Figure 7.7. Total offline selection acceptances for $H \to \gamma\gamma$. Three tightnesses of BDT-based selection cuts are shown in dark blue (standard), light blue (tighter), and light green (looser). The acceptance of the preselection is shown in yellow, while the full standard analysis cuts are shown in orange.

Turning to the signal MC η_{SC} in figure 7.8a, clear differences can be seen between the acceptances of the standard analysis cuts and the BDT-based selections. For all three BDT-based selection cuts, the signal efficiency is relatively constant in η_{SC}. As with the MVA-HLT results shown in the last section, this proves that the BDT-based selection is able to keep great kinematic coverage. The preselection and standard anaylsis, in addition to being much lower throughout the entire η_{SC} range, have clear holes in efficiency at the low-η regions of the endcap around $|\eta_{SC}| = 1.6$. This mirrors the behavior of the sHLT seen in section 7.2.1; it has been found that the hard R_9 cuts applied by both the sHLT and the preselection considerably decrease the signal efficiency in this region for both standard selections.

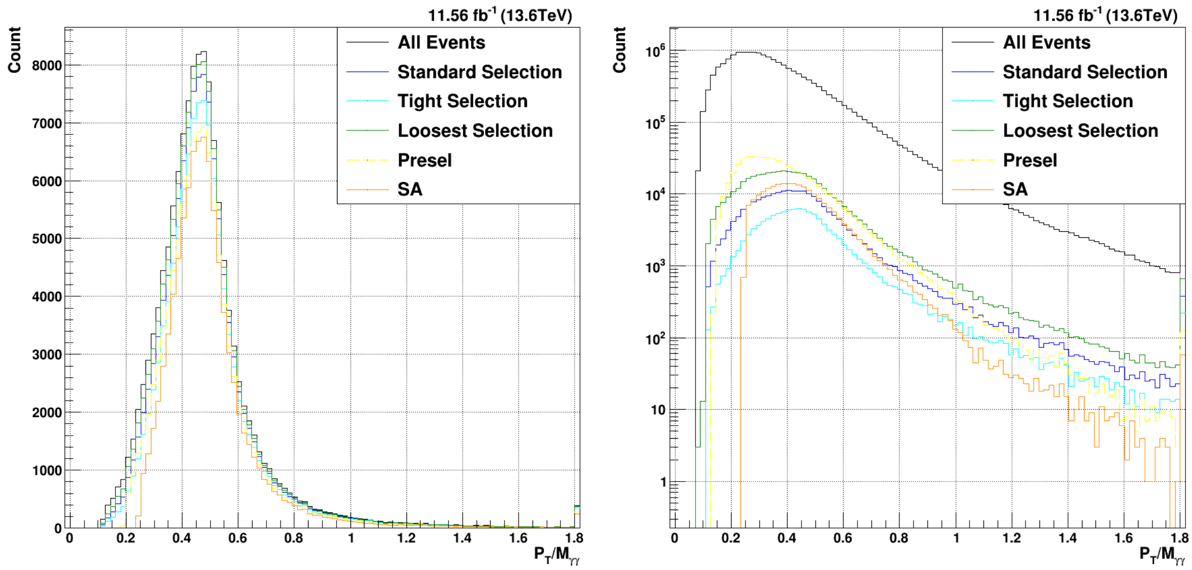
There are, additionally, clear differences with the data η_{SC} distributions shown in figure 7.8b. Comparing the standard BDT selection (blue) to the SA cuts (orange), it is immediately clear that the overall acceptance in the barrel detector is relatively equivalent. While the SA cuts seem to cut out slightly more data events at the very ends of the barrel near $|\eta_{SC}| = 1.4$, both selections shown are relatively flat in the barrel. In the endcap, however, some clear distinctions arise. While the SA cuts are relatively flat in the endcap, the BDT-based selections strongly favor the low-η regions of the endcap. Since the diphoton BDT for $H \to \gamma\gamma$ was trained using these signal MC and data distributions, it is expected that the data accepted will approximately follow the shape of the signal MC. Overall, however, this effect is not expected to have a major impact on any analysis.



(a) Signal MC η_{SC} **(b)** Data η_{SC}

Figure 7.8. Total offline selection acceptances for $H \to \gamma\gamma$. Three tightnesses of BDT-based selection cuts are shown in dark blue (standard), light blue (tighter), and light green (looser). The acceptance of the preselection is shown in yellow, while the full standard analysis cuts are shown in orange. Both the lead and sublead photons are plotted together.

The final distribution to consider, as with the HLT results, is the $p_T/M_{\gamma\gamma}$ in figure 7.9. Given the hard cuts on $p_T/M_{\gamma\gamma}$, the SA cuts show a clear loss of signal efficiency at lower values below about $p_T/M_{\gamma\gamma} < 0.3$. This is one of the major ways this BDT-based analysis is able to increase both the signal efficiency and kinematic coverage relative to the standard selection cuts. By instead accepting signal MC events all the way down to $p_T/M_{\gamma\gamma} = 0.1$, many events that contain a second photon with good $p_T/M_{\gamma\gamma}$ can be preserved for the analysis. Considering the background shown in figure 7.9b, it becomes clear why the $p_T/M_{\gamma\gamma}$ cuts were chosen; many data events fall below the selected thresholds, which makes this selection very effective at reducing the data acceptance. It is clear, however, that the BDT-based approach is still able to reduce the data acceptance to the desired rate while not making stringent requirements on $p_T/M_{\gamma\gamma}$.



(a) Signal MC $p_T/M_{\gamma\gamma}$ **(b)** Data $p_T/M_{\gamma\gamma}$

Figure 7.9. Total offline selection acceptances for $H \to \gamma\gamma$. Three tightnesses of BDT-based selection cuts are shown in dark blue (standard), light blue (tighter), and light green (looser). The acceptance of the preselection is shown in yellow, while the full standard analysis cuts are shown in orange. Both the lead and sublead photons are plotted together.

As with the HLT results, a table of signal efficiencies and data acceptances for both offline selections can be seen in table 7.3. As already discussed, the offline selection rates for both signal and data are considerably lower than those at the trigger level. This effect is much larger for the SA cuts, however. Whereas the BDT-based analysis only removes about 6% of the signal (91.17% vs 97.32% for the MVA-HLT), the standard analysis cuts remove about 18% of the signal MC (70.53 % vs 88.68% for the sHLT). The losses of the SA cuts are most pronounced in events where one photon is in each detector region. Here, the BDT-based analysis is able to increase the signal efficiency by a relative 63.9% (85.43% vs 52.12%). This further bears out the fact that the BDT-based analysis increases both the overall signal efficiency and the kinematic coverage of the offline selection. Taken together, it is clear that the MVA-HLT in conjunction with a BDT-based offline selection designed to cover it is a major step towards acheiving maximum signal efficiency while still being able to control the data acceptance rate.

Table 7.3. $H \to \gamma\gamma$ signal efficiencies and background acceptances for both offline selections, broken down by the location of each photon candidate in η_{sc}

(a) Signal Efficiencies

Region	BDT(%)	SA(%)	Gain(%)
B-B	94.56	81.95	+15.4
B-E	85.43	52.12	+63.9
E-E	91.63	63.63	+44.0
All	91.27	70.53	+29.4

(b) Data Acceptances

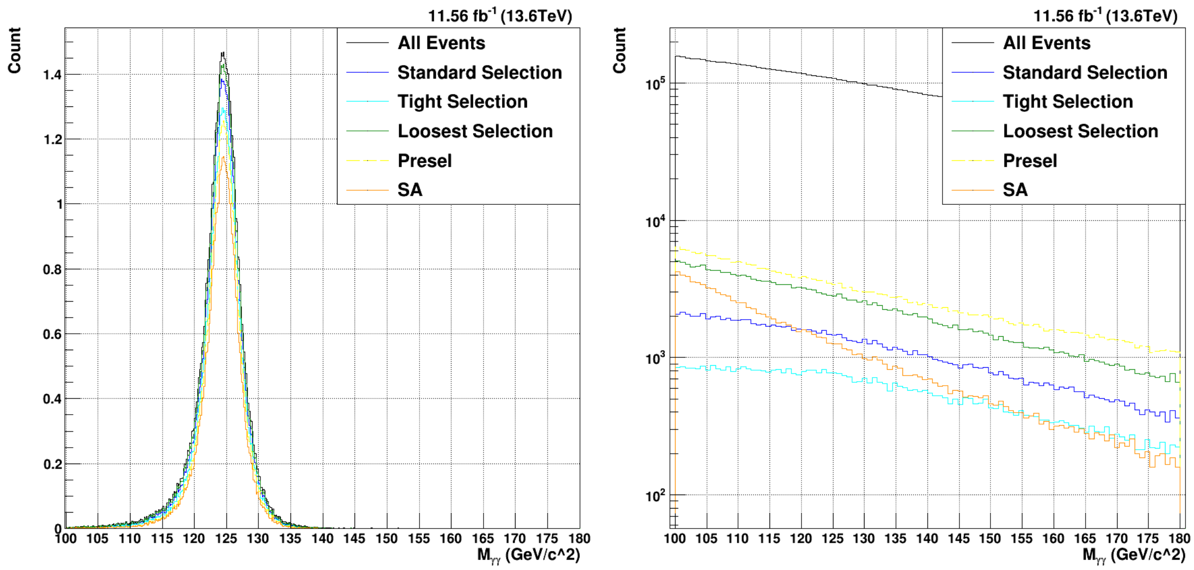
Region	BDT(%)	SA(%)	Gain(%)
B-B	2.50	2.30	+8.7
B-E	0.77	0.80	-3.8
E-E	1.08	1.71	-36.8
All	1.25	1.25	0.0

7.3.2 $HH \to b\bar{b}\gamma\gamma$ Offline Selection Acceptance

Of course, as has been discussed numerous tines, the main power in the BDT-based approach lies in its ability to increase the signal efficiency for rarer processes. While it is clearly always best to maximize the accepted signal, a process like $H \to \gamma\gamma$ will likely have enough accepted signal events to establish statistical significance in any case. Improvements in signal acceptance are, however, vital for low-rate processes such as $HH \to b\bar{b}\gamma\gamma$. Since there are expected to only be on the order of thousands (CHECK) of these events in the entirety of run 3, every single event erroneously discarded represents a major missed opportunity to establish statistical significance. In figures 7.10-7.12, as well as table 7.4, the improvements to signal efficiency and kinematic coverage will be examined for the gluon fusion $HH \to b\bar{b}\gamma\gamma$.

Many of the features seen in the $H \to \gamma\gamma$ diphoton mass distributions are still seen for the diHiggs sample (figure 7.10). The same improvement of signal acceptance can be seen, and similar differences in shaping of the background mass distribution between the different selections. There are, of course, a few numerical differences. Firstly, the SA efficiency for diHiggs signal MC is actually marginally lower than that of the single Higgs (67.78% vs 70.53%). Conversely, the BDT-based offline selection actually sees increased efficiency over that of the single Higgs (94.68% vs 91.27%). This leads to a total relative gain in signal MC efficiency of

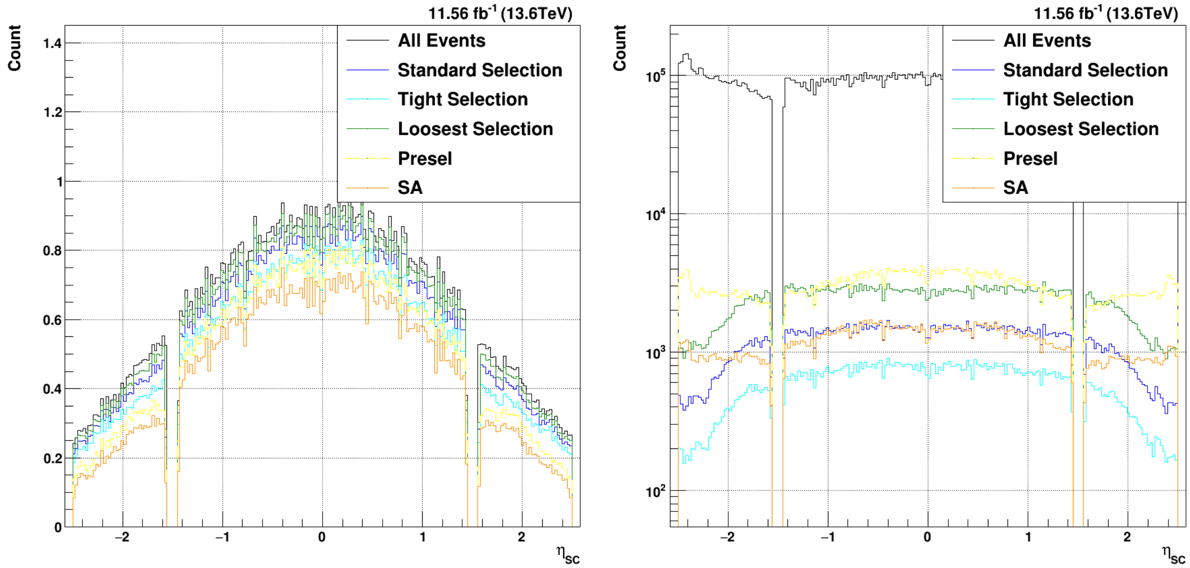
39.69%. This major gain in signal efficiency will substantively improve future diHiggs analyses. As with the $H \to \gamma\gamma$ signal, the width of the BDT-based analysis is marginally larger compared to that of the SA cuts ($2\sigma_{Eff.} = 4.55GeV/c^2$ vs $2\sigma_{Eff.} = 4.06GeV/c^2$).



(a) Signal MC $M_{\gamma\gamma}$ **(b)** Data $M_{\gamma\gamma}$

Figure 7.10. Total offline selection acceptances for $HH \to b\bar{b}\gamma\gamma$. Three tightnesses of BDT-based selection cuts are shown in dark blue (standard), light blue (tighter), and light green (looser). The acceptance of the preselection is shown in yellow, while the full standard analysis cuts are shown in orange.

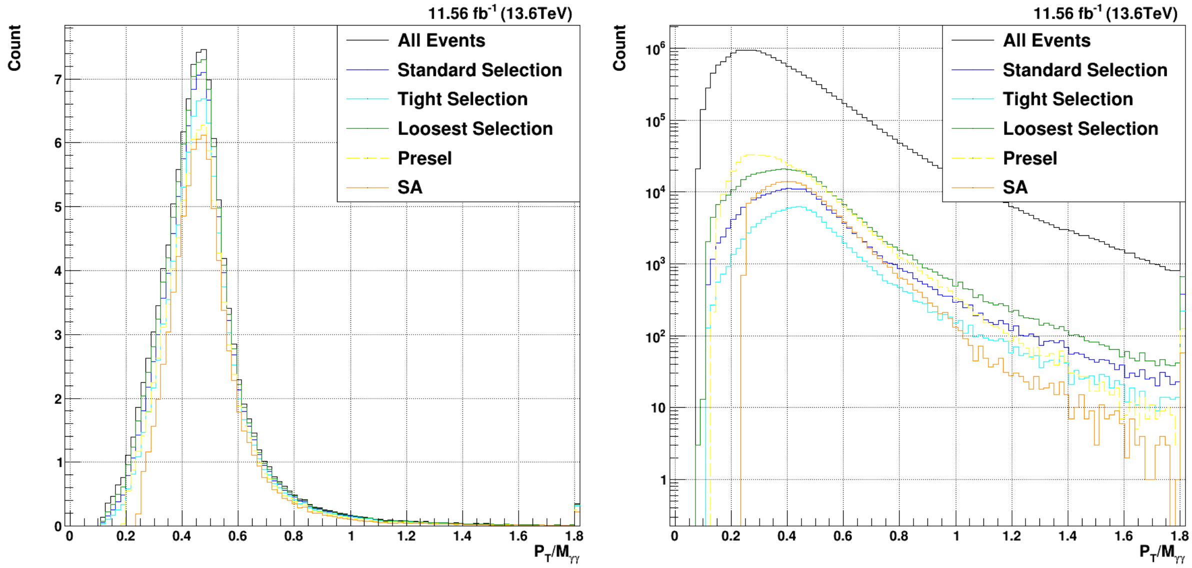
The η_{SC} distributions in figure 7.11 also show similar features to those of the single Higgs. Though the overall proportion of the diHiggs in the endcaps is lower, there is still an enhancement in the kinematic coverage around the low-η boundary of the endcap relative to the SA cuts. All three of the BDT-based selections keep consistent efficiency through the whole selected range of η_{SC}. The data distributions also mirror those of the single Higgs; while the SA cuts show relatively flat acceptance across the whole ECal, the BDT-based selections prefer more central events and cut out photons near the extreme η values.



(a) Signal MC η_{SC} **(b)** Data η_{SC}

Figure 7.11. Total offline selection acceptances for $HH \to b\bar{b}\gamma\gamma$. Three tightnesses of BDT-based selection cuts are shown in dark blue (standard), light blue (tighter), and light green (looser). The acceptance of the preselection is shown in yellow, while the full standard analysis cuts are shown in orange. Both the lead and sublead photons are plotted together.

The $p_T/M_{\gamma\gamma}$ distributions in figure 7.12 show similar features to those of the single Higgs. The loss of signal efficiency from the explicit $p_T/M_{\gamma\gamma}$ cuts can cleary be seen in figure 7.12a, while the power of these cuts to reduce the background acceptance can also be seen in figure 7.12b.



(a) Signal MC $p_T/M_{\gamma\gamma}$

(b) Data $p_T/M_{\gamma\gamma}$

Figure 7.12. Total offline selection acceptances for $HH \to b\bar{b}\gamma\gamma$. Three tightnesses of BDT-based selection cuts are shown in dark blue (standard), light blue (tighter), and light green (looser). The acceptance of the preselection is shown in yellow, while the full standard analysis cuts are shown in orange. Both the lead and sublead photons are plotted together.

The overall $HH \to b\bar{b}\gamma\gamma$ signal MC efficiencies and data acceptances can be seen in table 7.4. As with the single Higgs sample, considerable gains are measured accross the whole detector. A particular enhancement is still seen in B-E events, where a relative 64.2% gain is seen. These massive gains in both the kinematic coverage and the overall signal efficiency should prove a major boon to future diHiggs analyses, and demonstrate the possible power of leveraging the MVA-HLT for future analyses.

Table 7.4. $HH \to b\bar{b}\gamma\gamma$ signal efficiencies and background acceptances for both offline selections, broken down by the location of each photon candidate in η_{sc}

(a) Signal Efficiencies

Region	BDT(%)	SA(%)	Gain(%)
B-B	97.23	73.43	+32.4
B-E	88.38	53.83	+64.2
E-E	89.88	57.02	+57.6
All	94.68	67.78	+39.7

(b) Data Acceptances

Region	BDT(%)	SA(%)	Gain(%)
B-B	3.12	2.42	+28.9
B-E	0.76	0.86	-11.6
E-E	1.50	1.71	-12.3
All	1.46	1.46	0.0